INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, November - December, 2025

Degree: **B.C.A** Semester: **First**

Stream: **BCA**

Paper Code: **BCACC101**

Paper Name: **Introduction to C Programming**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

1. Find errors, if any, in the following program: 2

```
#include (stdio.h)
void main (void)
{
    print ("Hello C");
}
```

2. Why and when do we use the **#define** directive? 2

3. Where are blank spaces permitted in a C program? 2

4. State the differences between the declaration of a variable and the definition of a symbolic name. 2

5. What is printed by the following program? 2

```
main
{
   int x = 5, y = 10, z = 10;
   x = y = = z;
   printf("%d",x) ;
}
```

6. What is the purpose of **scanf()** function? 2

7. What precautions should be taken while using **scanf()**? 2

8. What will be the output of the following code? 2

```
char s1[ ] = "ANIL KUMAR GUPTA";
char s2[ ] = "KUMAR";
printf (strstr (s1, s2) );
```

9. Enumerate the rules that apply to a function call. 2

10. Write a function that will round a floating-point number to an indicated decimal place. For example the number 17.457 would yield the value 17.46 when it is rounded off to two decimal places. 2

11. Explain the syntax and use of **if, if-else**, and **nested if-else** statements. 2

12. What is the use of **break** and **continue** statements? 2

13. What are the advantages of using functions? 2

14. How can you compare two strings in C? 2

15. What are function prototypes? Why are they used? 2

Part - B

Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Relationship between Celsius and Fahrenheit is governed by the formula 5

$$F = \frac{9C}{5} + 32$$

Write a program to convert the temperature
a) from Celsius to Fahrenheit and
b) from Fahrenheit to Celsius

17. Write a program that will obtain the length and width of a rectangle from the user and compute its area and perimeter. 5

18. What are constants in C? Describe different types with examples. 5

19. What is a dynamic array? How is it created? Give a typical example of use of a dynamic array. 5

20. Character strings in C are automatically terminated by the null character. Explain how this feature helps in string manipulations. 5

21. Distinguish between the following: 5
(a) Actual and formal arguments
(b) Global and local variables
(c) Automatic and static variables
(d) Scope and visibility of variables
(e) & operator and * operator

22. How does an append mode differ from a write mode? 5

23. Enumerate the differences between functions and parameterized macros. 5

24. Write a program to find the largest of three numbers using **if-else.** 5

Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Write a program to count and print the number of negative and positive numbers in a given set of numbers. Test your program with a suitable set of numbers. Use scanf to read the numbers. Reading should be terminated when the value 0 is encountered. 10

26. Distinguish between the following pairs: 10
(a) getchar and scanf functions.
(b) %s and %c specifications for reading.
(c) %s and %[] specifications for reading.
(d) %g and %f specification for printing.
(e) %f and %e specifications for printing.

27. Find errors, if any, in the following function definitions: 10

(a)
```
void abc (int a, int b)
{
        int  c;
        .....
        return (c);
}
```

(b)
```
int abc (int a, int b)
{
        ....
        ....
}
```

(c) int abc (int a, int b)

```
{
        double c = a + b;

        return (c);
}
(d) void abc (void)
{
        . . . .
        . . . .
        return;
}
```

28. Develop a modular interactive program using functions that reads the values of three sides of a triangle and displays either its area or its perimeter as per the request of the user. Given the three sides a, b and c. **10**

Perimeter = a + b + c

Area = $\sqrt{(s-a)(s-b)(s-c)}$

where, s = (a + b + c)/2

29. Define a structure data type named **date** containing three integer members **day, month** and **year**. Develop an interactive modular program to perform the following tasks; **10**

a) To read data into structure members by a function.

b) To validate the date entered by another function.

c) To print the date in the format April 29, 2002 by a third function.

The input data should be three integers like 29, 4, and 2002 corresponding to day, month and year. Examples of invalid data:

31, 4, 2002 - April has only 30 days

29, 2, 2002 - 2002 is not a leap year

30. Write a function **day_name** that receives a number n and returns a pointer to a character string containing the name of the corresponding day. The day names should be kept in a static table of character strings local to the function. **10**

31. Write a program to check whether a year is a leap year or not. **10**

32. Write a program to concatenate two strings without using library functions. **10**

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